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Theory of Computation

Submission link: https://forms.gle/9QfcWiqewGmCcvG36

Exercise 6:

(Pumping Lemma)

Prove by Pumping Lemma that the following languages are not regular.

1. L1 = $\{ 0^i 1^j : j \text{ is a multiple of } i \}$

Proof: Assume that L_1 is regular. Let m be the number of states in DFA.

We pick a string $w \in L_1$, $|w| \geq m$

Decompose w into xyz; $|xy| \leq m$, $|y| \geq 1$

$\overbrace{0\,0 \ldots 0}^{m}\overbrace{1\,1\,1 \ldots 1}^{m}$; $y = 0^k$; $k \geq 1$

$|-x-||-y-||-z-|$

We pick $i = 2$, $xy^2z \Rightarrow xyyz = 0^{m+k}1^m \notin L_1$ $\quad \overbrace{m+k}^{i} > \overbrace{m}^{j}$

Since m is not a multiple of $m+k$, $0^{m+k}1^m \notin L_1$.

Contradiction with P.L. ∴ L_1 is not Regular

2. $L2 = \{ 0^i 1^j : i < j \}$

Assume that L_2 is regular. Let m be # of states in DFA

We pick a string $w \in L_2$. $|w| \geq m$. $w = 0^m 1^{m+1}$

Decompose w into xyz, $|xy| \leq m$, $|y| \geq 1$

$$\overbrace{\underbrace{0}_{x}\underbrace{0\ldots0}_{y}}^{m}\overbrace{\underbrace{11\ldots1}_{z}}^{m+1} \quad ; \; y = 0^k ; \; k \geq 1$$

We pick $i=2$, $xy^2z = xyyz = 0^{m+k}1^{m+1}$

Since $m+k \geq m+1$, $0^{m+k}1^{m+1} \notin L_2$

Contradiction with P.L.! $\therefore L_2$ is not regular

*3. L3 $=\{ 0^i 1^j : i \geq j \}$

Assume that L_3 is regular. Let m be the no. of states in DFA

We pick a string $w \in L_3$, $|w| \geq m$

$w = 0^m 1^m$

Decompose w into xyz, $|xy| \leq m$, $|y| \geq 1$

$\underbrace{\overbrace{00\ldots0}^{m}}_{x\,y}\,\underbrace{\overbrace{11\ldots1}^{m}}_{z}$; $y = 0^k$; $k \geq 1$

We pick $i=0$ to pump xyz

Since $m-k < m$, $0^{m-k}1^m \notin L_3$

Contradiction with P.L.! $\therefore L_3$ is not regular